Instructions for Annotations

The primary objective of this annotation exercise is to create a labeled corpus of participant reading recordings.

Model Training: To train Automatic Speech Recognition (ASR) models to transcribe recordings of adults across all Indian states reading short passages and word-lists. The transcript generated by the ASR will be compared with the reference text to assess the reading.

Model Evaluation: To create a ground truth transcript which can be used to evaluate the ASR models.

Crucial Note:

- **The transcription must in verbatim**
- **Annotators must NOT correct grammar, fix pronunciation, fill in intended words, or infer meaning.**

Inputs Provided to the Annotator

The annotator will be provided with the following required elements for each annotation task:

1. **Audio File:** A recording of a student reading (either a list of words or a paragraph).
2. **Reference Text:** The exact **Word List** or **Spoken Paragraph** the student was instructed to read. This text will be used to identify miscues.
3. **Sample Adult voice recordings:** Recordings of a few reference texts provided by the state stakeholder for accent reference.

Protocol

Core Principle: **Transcribe the whole audio as heard while respecting accent variations.**

1. The whole speech in audio needs to be transcribed irrespective of what the reference text is.

2. The speech needs to be transcribed as heard.

The following variations can be tolerated in pronunciation –
a. Plausible variation in syllables stressed within a particular word. For example –
i. “de-VEL-op-ment” and “DE-vel-op-ment”
ii. “de-TER-mine” and “de-ter-MINE”
b. Plausible variation in syllable sounds based on the letters being read. For example,
i. Plausible variation – The word “Education” can be pronounced as
“E-du-kay-shon” (Standard) or “E-joo-kay-shun” (Variation)
ii. Implausible variation – The word “Education” pronounced as
“E-zu-ka-shon” (Not a common pronunciation)

3. In case of ambiguity regarding the pronunciation of a word, the UK, US or Indian English pronunciations in Google can be referred to as “tie-breakers”.

4. If found correct, the transcript for the word must be the same as the word in the reference text. Otherwise, the word needs to be transcribed as heard (verbatim).

Steps :

1. **Timestamping**
   - Mark the **start and end time of every spoken word** using the same time format (e.g., **`HH:MM:SS.MS`**).
   - Timestamps should be **as accurate as possible**, closely matching when the word actually starts and ends.
   - If two words are spoken together and it’s hard to find an exact boundary, **estimate a reasonable split**.
   - **Background noise** must also be timestamped, with a start and end time.
   - **Pauses do not need timestamps**.
   - **Filler sounds** (like umm, uh) should be timestamped and enclosed within the appropriate **filler tags**.

2. **Transcription of recognisable (known) English words matching the reference word**
   - If the student pronounces a word **correctly**, transcribe it **exactly as it appears in the reference text**.
   - A pronunciation is considered **correct** if it matches **any accepted US, UK, or Indian English pronunciation**, or the provided reference audio.
   - **Do not change spelling** based on how it sounds. Always **copy-paste the reference word**.
   - Different acceptable accents or pronunciations are **all treated as correct**.
   - **Example:**
     I. *leisure* pronounced as *leh-zhuh* or *lee-zher* → transcribe as **leisure**
     II. *schedule* pronounced in American, British, or Indian style → transcribe as **schedule**
     III. *stove* pronounced close to the reference audio → transcribe as **stove**, not based on sound alone

3. Transcription of recognisable (known) English words not matching the reference word

- If the student says a **real English word**, but it is **different from the reference word**, transcribe **what was actually said**, not the reference word.
- The pronunciation is considered **incorrect** if it does **not match** accepted US, UK, or Indian pronunciations of the **reference word**, or the provided reference audio.
- Write the word in **Indian English spelling**, **not in Devanagari**.
- **Do not correct** the word to match the reference text.
- **Example:**

  I. *colon* pronounced as *kuh-lur* → transcribe as **colour**
  II. *man* pronounced as *main* → transcribe as **main**
  III. Reference word *house*, spoken as *horse* → transcribe as **horse**

4. Transcription of invalid (not actual) English words

- If the spoken word is **not a real English word**, transcribe **exactly how it sounds**.
- Use the **Devanagari script** to capture the pronunciation.
- The transcription **does not need to be a real Hindi word**.
- Focus on **accurate phonetic representation**, including vowels, matras, and halant where needed.
- Do **not try to correct or guess** the intended English word.
- **Example:**

  I. *pigeon* pronounced as *pid-uh-jon* → **पिडजोन**
  II. *literacy* spoken as *lit-ruh-sary* → **लिट्र सरी**
  III. *blak* (short 'a' sound) → **ब्लक**
  IV. *sheih-p* ending abruptly on *p* → **शाइप्**

5. Transcription of words spoken as letter names

- If a word is **spelled out letter by letter**, transcribe **each letter name** as it is spoken.
- Use the **Devanagari script** for letter names, since they are **not full English words**.
- Transcribe **exactly what is heard**, even if the letter pronunciation is non-standard.
- Use your **best judgment** when letter sounds differ from standard pronunciations.
- **Example:**

  I. *balloon* read as *b-a-l-l-o-o-n* → **बी ए एल एल ओ ओ एन**
  II. *pan* read as *pee-yay-yen* → **पी ये येन** (not **पी ए एन**)

6. Transcription of recognisable English words spoken sub-lexically with pauses

- If a word is spoken in **parts with pauses in between**, transcribe **each part separately**.
- Decide whether to use **English or Devanagari** for each part:

  I. Use **English** if the part is a **real English word** with correct pronunciation.
  II. Use **Devanagari** if the part is **not a real English word**.

- If a word is spoken in **parts with pauses in between**, transcribe **each part separately**.
- **Example:**

  I. *compute* spoken as *com … pute* → **कं प्यटू**
  II. *prolong* spoken as *pro … long* → **pro long**
  III. *vineyard* spoken as *vin … yard* → **विन yard**

7. Transcription of recognisable English words spoken sub-lexically with stretches

- If a word is spoken with **stretched sounds** (elongated vowels or consonants), transcribe the **entire word in Devanagari**.
- Do **not split** the word into parts.
- To show stretching, add **only one extra vowel or consonant**, no matter how long the stretch is.
- Focus on representing the **sound**, not correct spelling.
- **Example:**

  I. *coming* spoken as *co…ming…* → **कअमिगंअ**

8. Transcription of proper nouns

- Names of **people, places, animals, or other named entities** should be transcribed in the **Devanagari script**.
- Transcribe based on **how the name is pronounced**, not English spelling.
- Do **not correct or anglicize** the pronunciation.
- **Example:**

  I. *Karthik / Kartik* → **कार्तिक**
  II. *England* (standard pronunciation) → **इंग्लेंड**

9. Transcription for false starts and repetitions

- Transcribe **everything exactly as spoken**.
- Include **false starts, stutters, and repetitions**.

- Follow the same transcription rules (English or Devanagari) based on **how each part is spoken**.
- Do **not remove or correct** repeated or restarted sounds.
- **Example:**

  I. *fire* read as *f-i-r-e* and then correctly → **एफ़ आई आर ई fire**

  II. *clown* spoken as *c-c-clown* → **क क clown**

10. Transcription of punctuations within a word

i) Punctuations within a word are to be included. A few examples of punctuations within words are :
"You're" (contraction), "Dog's home OR Dogs' homes" (possessive), "catch-up" (hyphenated)

Examples:
(1) Reference text: "don't"; what is heard: "do not"; annotation: "do not".
(2) Reference text: "Who's"; what is heard:who's; annotation: "who's".
(3) Reference text: "who's"; what is heard: "हस"; Annotation: "हस"

(note that the transcript must be in Devanagari as हस is not a valid English word)
(4) Reference text: "who's"; what is heard: "has"; Annotation: "has".

11. **Abbreviations**.

If the abbreviation sounds like an English word, transcribe it in English, otherwise use Devanagari.

i) Examples:
(1) If "DVD" is read in the standard manner, transcribe it in Hindi as डीवीडी. If you hear pauses while saying the individual letters, add spaces as well.
(2) If "SQL" is read as "sequel", transcribe it as "sequel".
(3) If ISRO is read as "isro" transcribe it in Devanagari इसरो

10. Labels

Category	Explanation	Label	Reference Text (Example)	Transcription (Example)
Unintelligible speech/mumbling	When the speech is indiscernible and is not transcribable either in English or in Devanagari.	<MB>	There is a small pond. Ducks swim in the pond. A frog jumps into the pond. I see fish in the pond. The pond water is very cool.	After "there is" all unintelligible, there is <MB>

Noise	When there is ambient background noise, or if other speech can be heard (not of the child).	<NOISE>	NA	NA
Spelling out words with letter names	When a word is spelled out with letter names (see letter case rules above).	<LN>	cat, pin, sun, map, stop, plan, trip, milk, hand, fast, bake, face, cab, large, game	**<LN> पी आई एन <LN/>** **<LN> एस यू एन <LN/>** **<LN> ए म ए <LN/>**
Filler sounds	When there are filler words like “aaah”, “ummm”, transcribe as heard in Devanagari and add the <FIL> tag to each word.	<FIL>	NA	NA
Silences	Long pauses or silences (> 2 seconds).	<SIL>	NA	NA

11. Output format


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            "end": "0:00:01.209974",
            "Transcription": [ "Knowing" ]
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        {
            "start": "0:00:01.539350",
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            ”Transcription”: [ ”<NS> जोम्प् </NS>” ]
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"end": "0:00:04.651974",

”Transcription”: [”स्लीड”]

} ,

{

"start": "0:00:05.232074",

"end": "0:00:09.651974",

”Transcription”: [“<LN> बी

ए

एल्

एल् </LN>]

}

]

}